

PRINCE ALBERT, SK, Canada

WMO: 718690

Lat: 53.2103N

Lon: 105.6767W

Elev: 1405

StdP: 13.97

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-25.7	-38.8	0.6	-30.7	-33.6	0.9	-25.4	23.5	15.4	20.5	15.5	3.4	260	0.683

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	83.8	66.3	80.6	64.7	77.8	62.8	68.8	79.9	66.7	77.3	64.8	74.4	10.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	96.9	74.2	62.8	89.9	71.5	61.0	84.5	69.6	33.9	80.6	32.1	77.3	30.6	74.5	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	18.6	16.7	-38.4	90.3	5.9	3.5	-42.7	92.8	-46.1	94.8	-49.5	96.8	-53.8	99.3
			-38.5	72.9	5.9	2.0	-42.7	74.3	-46.2	75.5	-49.5	76.5	-53.7	78.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	1.9	5.9	18.8	36.9	50.2	59.6	64.5	62.2	52.4	38.2	20.6	6.4
	DBStd	24.82	15.76	14.16	14.18	10.52	8.38	6.30	5.16	5.94	7.81	8.40	11.99	14.26
	HDD50	6877	1490	1235	966	405	101	6	0	1	63	376	882	1352
	HDD65	11085	1955	1655	1431	844	460	182	71	126	381	831	1332	1817
	CDD50	1388	0	0	0	11	109	293	451	378	136	10	0	0
	CDD65	120	0	0	0	0	3	19	57	38	3	0	0	0
	CDH74	1638	0	0	0	6	105	290	614	504	116	3	0	0
	CDH80	340	0	0	0	1	19	61	117	120	22	0	0	0
Wind	WSAvg	7.4	6.6	6.6	7.6	8.6	8.8	8.0	7.1	6.9	7.4	7.6	7.2	6.5
Precipitation	PrecAvg	16.8	0.7	0.6	0.7	1.0	1.7	2.9	3.0	2.2	1.5	1.0	0.7	0.8
	PrecMax	23.6	1.4	1.8	2.2	2.6	5.9	6.4	7.0	7.8	4.2	3.6	1.7	1.6
	PrecMin	11.0	0.1	0.0	0.1	0.0	0.2	0.3	0.8	0.5	0.1	0.0	0.1	0.3
	PrecStd	2.9	0.3	0.4	0.4	0.7	1.3	1.5	1.5	1.6	1.0	0.8	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.0	40.9	51.1	71.8	82.4	86.3	87.4	88.2	83.4	70.5	50.7	40.8
		MCWB	35.3	35.1	42.3	52.1	59.1	65.3	69.5	66.3	64.3	55.7	42.5	35.9
	2%	DB	35.0	35.1	45.3	65.2	76.6	80.8	83.1	83.1	77.2	62.0	44.1	35.2
		MCWB	31.9	31.3	38.2	49.0	56.7	63.0	67.9	66.0	59.9	50.2	38.7	31.9
	5%	DB	29.9	30.7	41.4	60.0	72.3	77.1	80.3	79.4	72.3	56.8	39.5	29.7
		MCWB	27.8	28.2	35.8	46.8	54.6	60.7	65.9	64.5	58.2	47.2	35.6	27.7
	10%	DB	24.8	26.7	37.5	54.4	68.2	73.5	77.4	76.1	67.5	51.8	35.7	26.1
		MCWB	23.4	24.6	33.5	43.7	52.7	59.2	64.4	62.9	55.6	45.1	32.9	24.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.4	35.6	42.7	53.6	61.8	68.8	72.8	70.5	65.6	56.6	43.5	36.4
		MCDB	39.3	40.3	50.1	68.0	74.9	81.5	82.8	82.2	78.8	69.3	50.0	40.4
	2%	WB	32.1	31.7	38.6	50.4	58.7	65.3	69.9	67.8	62.0	51.8	39.2	31.8
		MCDB	34.9	34.7	44.4	62.6	73.0	76.0	80.6	79.5	73.5	58.9	43.4	34.8
	5%	WB	27.9	28.3	36.1	47.5	56.1	63.2	67.6	65.7	59.1	48.2	35.9	28.0
		MCDB	29.9	30.7	40.9	58.8	69.9	73.0	77.6	76.8	70.5	56.1	39.1	29.8
	10%	WB	23.3	24.6	33.6	44.4	53.6	61.2	65.7	63.9	56.5	45.1	33.0	24.5
		MCDB	24.7	26.7	37.2	53.3	65.9	70.2	75.0	73.5	66.1	50.9	35.7	26.1
Mean Daily Temperature Range	5% DB	MDBR	18.4	20.4	19.8	21.1	24.3	21.2	20.9	22.2	22.2	18.0	14.7	16.3
		MCDBR	22.3	22.3	21.9	30.7	31.7	28.1	26.1	28.7	31.8	28.1	19.6	20.1
		MCWBR	19.6	18.9	15.8	17.4	15.6	13.3	13.3	14.0	16.7	17.4	14.7	17.7
	5% WB	MCDBR	21.7	21.7	21.0	28.5	28.3	22.4	23.6	25.2	29.1	26.8	18.7	19.3
		MCWBR	19.2	18.4	15.3	17.1	14.6	12.2	13.1	13.1	15.9	17.3	14.4	17.1
Clear-Sky Solar Irradiance	taub	0.234	0.250	0.301	0.336	0.354	0.375	0.376	0.369	0.317	0.288	0.248	0.226	
	taud	2.319	2.316	2.265	2.360	2.367	2.351	2.336	2.363	2.508	2.542	2.453	2.331	
	Ebn at Noon	247	278	281	285	284	278	275	270	275	263	243	229	
	Edh at Noon	18	25	33	34	36	37	37	34	26	20	16	15	
All-Sky Solar Radiation	RadAvg	316	653	1152	1537	1820	1826	1884	1564	1098	629	290	220	
	RadStd	31	37	81	134	159	139	121	128	111	101	40	28	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.68	+1.10	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page